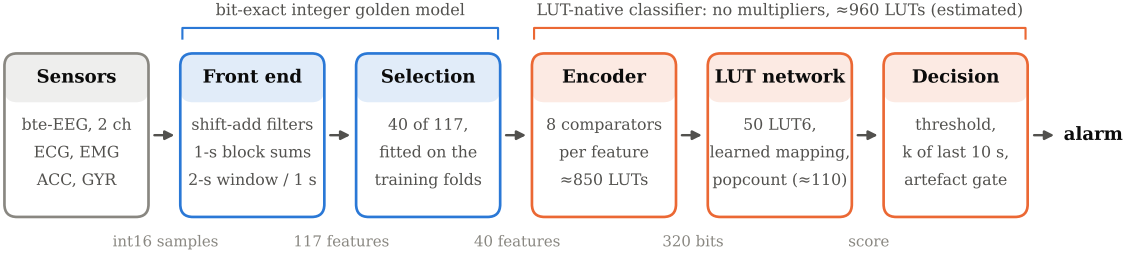
bit-exact integer golden model
LUT-native classifier: no multipliers, ≈960 LUTs (estimated)
Sensors
bte-EEG, 2 ch
ECG, EMG
ACC, GYR
Front end
shift-add filters
1-s block sums
2-s window / 1 s
Selection
40 of 117,
fitted on the
training folds
Encoder
8 comparators
per feature
≈850 LUTs
LUT network
50 LUT6,
learned mapping,
popcount (≈110)
Decision
threshold,
k of last 10 s,
artefact gate
alarm
int16 samples
117 features
40 features
320 bits
score